

Wireless Charging is the Future

Wireless charging is no more inferior to wired charging. Better standards, integration methods and components have evolved over the past few years to make wireless charging a better technology. And it need not be confined to charging cell phones. Find out how

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The multimedia-rich portable electronic devices available now end up consuming a lot of energy just to keep the device's graphics processing unit juiced up. Let us take a cell phone for instance. The traditional approach is to look for a wall socket and push in the wired charger's plug to charge the battery.

Though a cell phone can also be charged wirelessly, most people are wary of the time it takes to charge the phone wirelessly. No one wants to wait for hours to charge a phone's battery wirelessly, especially when you need to use the device in next 15-20 minutes. But does wireless charging still take a long time to power up a device?

Although called wireless, wireless charging is not totally without wires. The primary coil in the charger needs current from AC mains to produce magnetic flux, which produces current in the charger's secondary coil by induction for charging the device. So the primary coil is plugged into an AC socket using wires, though called wireless. Wireless charging, when started, was based on Qi inductive power standard that was produced in 2008 by WPC (Wireless Power Consortium).

The low-power Qi specification limits the draw of power to just 0W to 5W, which limits the current available to the device for charging. But newer wireless chargers available in the market are much better and can transmit a higher amount of current. A few years ago, Qi began to extend its specifica-

tion to medium power to deliver up to 120 watts.

Chips like Broadcom's BCM59350 let gadgets use any wireless charging standard. What's more, it is not strangled by the previous 5W limit and can go up to 50% higher levels without breaking sweat. This results in a charging speed that can finally compete

pad, maximum amount of power gets transferred to the device to charge it easily and fast.

The chargers based on A4WP standard transfer power wirelessly at a frequency of 6.78 MHz, which is internationally available for such applications. The frequency used for control and management is within the 2.4GHz band, which is ideal for use with smartphones and with other electronics devices globally.

Itay Sherman, CTO, Power Matter Alliance says, "Although both inductive and resonant charging are based on the same principle, the main difference between the two is the coupling factor. It defines how the two coils will be aligned to transfer maximum flux.

When measured on a scale 0 means less coupling factor and 1.0 means high coupling factor. Devices based on inductive charging technology have a coupling factor of 0.3 to 0.8 whereas devices based on resonance charging can also have a coupling factor below 0.3."

with what your travel charger gives you through its wire. Modern wireless charging allows transfer of upwards of 1A current.

Resonance charging

The new wireless charging standard A4WP (Alliance for Wireless Power) is changing the concept of inductive charging from magnetic induction to resonance. Magnetic induction requires the two coils to be placed very closely, whereas with this technique a large area is covered so the device's position is not very critical. In fact, multiple devices can be charged wirelessly with this technique by placing them all within the wide resonant field. And charging is not affected by the position of the device being charged. No matter how you position the device on the charging

Charger-receiver alignment

Quick and proficient charging is possible only when the primary coil (transmitter) identifies the secondary (receiver). If the transmitter works efficiently but the receiver does not, the wireless charger loses its significance.

Texas Instruments has recently come up with two 5W receivers (bq51020 and bq51021) focussed around the Qi 1.1 standard. They offer a programmable yield voltage of about 8V, reducing the power loss and the rise in temperature by 35% compared

